

EDUCATION PROJECT

Analysis

Report

FOUNDATIONS OF DATA SCIENCE, DATA 5100

PROBLEM STATEMENT

The purpose of the assignment was to perform work using the data science methodology to answer the question: **Does school geographical assignment have a strong effect on school performance?**

METHODOLOGY

Technique

To do answer this question, we will be analyzing the relationship between a student body's average ACT - and their equivalent SAT - scores and the schools' locale classification (city, suburban, town, rural).

We expect to see a range of ACT/SAT exam scores, but to accurately answer our question, we will also be analyzing the relationship of exam scores against five other socioeconomic factors - unemployment rate, percentage of adults with a college degree (higher-educational attainment), percentage of children in a married couple family, median household income, and percentage of students at the school that are eligible for free or reduced price lunch (poverty rate).

Although our problem statement focuses on one socioeconomic factor; we want to incorporate other factors into our analysis so that we can better understand **the level** at which geographical assignment affects school performance.

Approach

These conclusions will be decided based on what we find when working through our descriptive and regression analysis of the clean data.

The exploratory data analysis segment of this process will help us visualize the relationships between average ACT/SAT exam scores and all the socioeconomic variables, while the regression analysis segment will help us test these findings statistically.

Based on what we discover in these analyses, we will use a **Diagnostic Approach** to make inferences about the relationship between a school's geographical assignment and academic performance.

DATA COLLECTION

This project utilizes three main data sources.

The **first dataset**, *EdGap_data.xlsx* (EdGap.org, 2017), provides average ACT/SAT scores and five socioeconomic variables for U.S. high schools, including household income, unemployment, educational attainment, family structure, and free/reduced lunch percentages. Socioeconomic variables are based on the U.S. Census Bureau's American Community Survey, and ACT/SAT data were collected from state departments of education or other public data releases.

The **second dataset**, *ccd_sch_029_1617_w_1a_11212017.csv*, is from the National Center for Education Statistics (NCES) *Common Core of Data (CCD): Public School Universe Survey, 2016–2017*. This dataset provides identifying information for each school—such as ID, name, city, state, ZIP code, and school type—allowing linkage to EdGap's performance and socioeconomic data.

The **third dataset**, *EDGE_GEOCODE_PUBLICSCH_1617.xlsx*, is also published by NCES under the *Education Demographic and Geographic Estimates (EDGE)* program. This dataset provides latitude, longitude, and locale codes for schools, enabling each school to be categorized into one of four

geographic classifications (City, Suburban, Town, Rural) following the standards in the *EDGE Geocode Technical Documentation* (NCES, 2020).

Together, these datasets allow for the analysis of relationships between school performance, socioeconomic conditions, and geographical information across public high schools.

DATA ANALYSIS

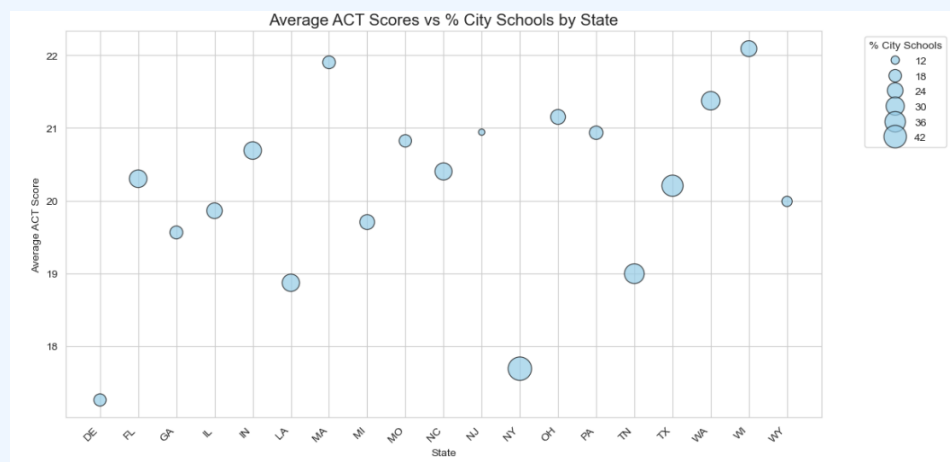
Exploratory Analysis, by Geographical Assignment

In this segment of our analysis process, we aimed to gain a descriptive understanding of the relationships between the ACT/SAT exam scores and all the variables.

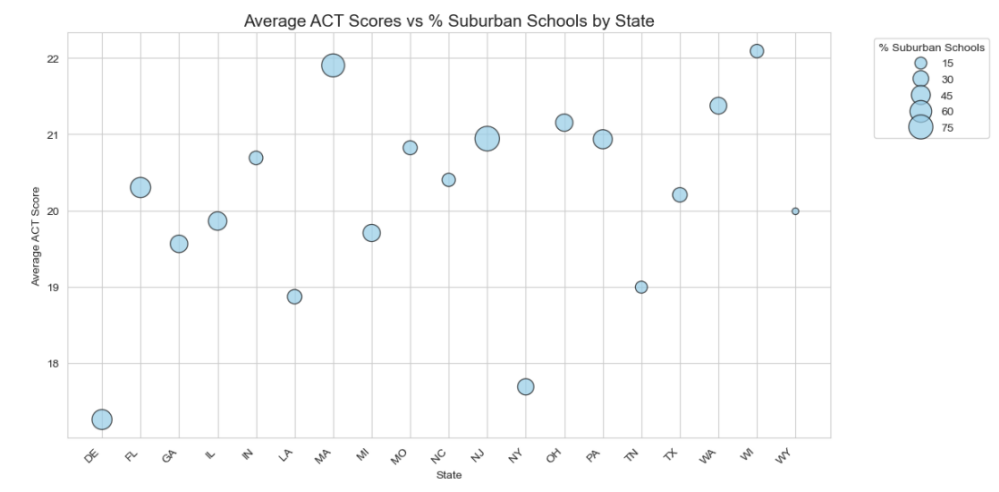
When analyzing average **ACT/SAT exam scores by geographical assignment**, we were able to determine that performance had very little variance across all four categories. For example, students who attended schools in a **Town and Rural** geoassignments scored approximately the same in the year of 2016; approximately 20 points. Conversely, the geoassignment category that displayed the highest exam scores were **Suburban** schools, scoring approximately 21 points on average, and the geoassignment category that displayed the lowest exam scores were City schools, scoring approximately 19 points on average.

We also analyzed these exam scores on a state level. Instead of analyzing average exam scores by school, we gain a different perspective when analyzing these relationships **by state representation**.

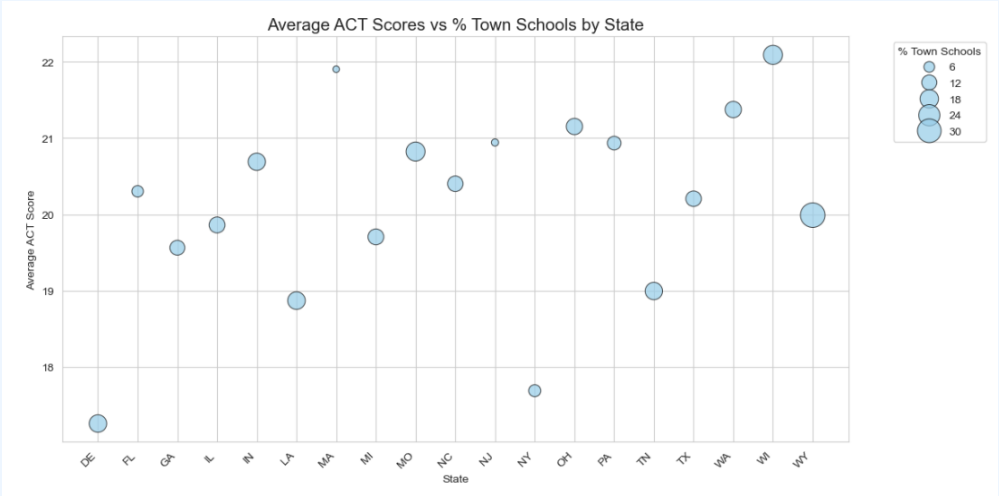
Percent of City Schools by State:



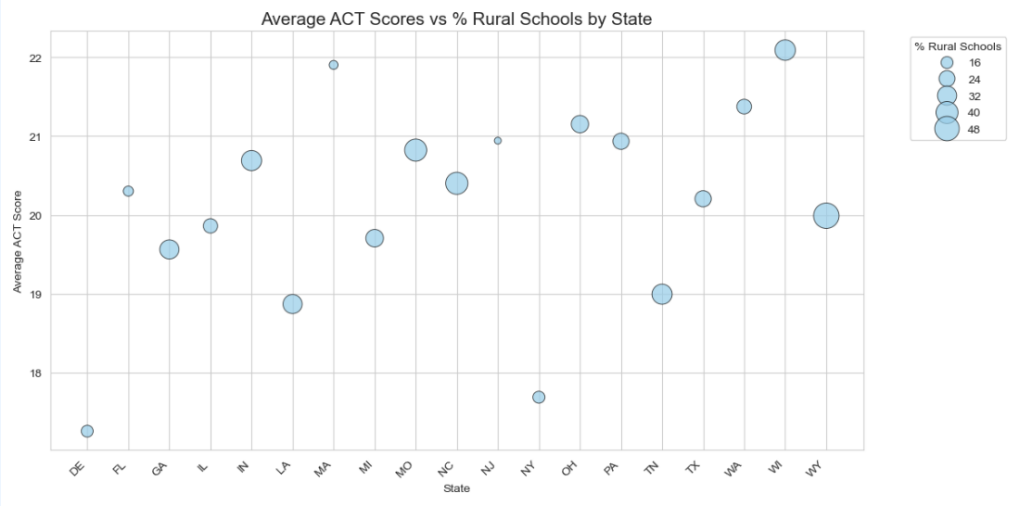
Percent of Suburban Schools by State:



Percent of Town schools by State:



Percent of Rural Schools by State:

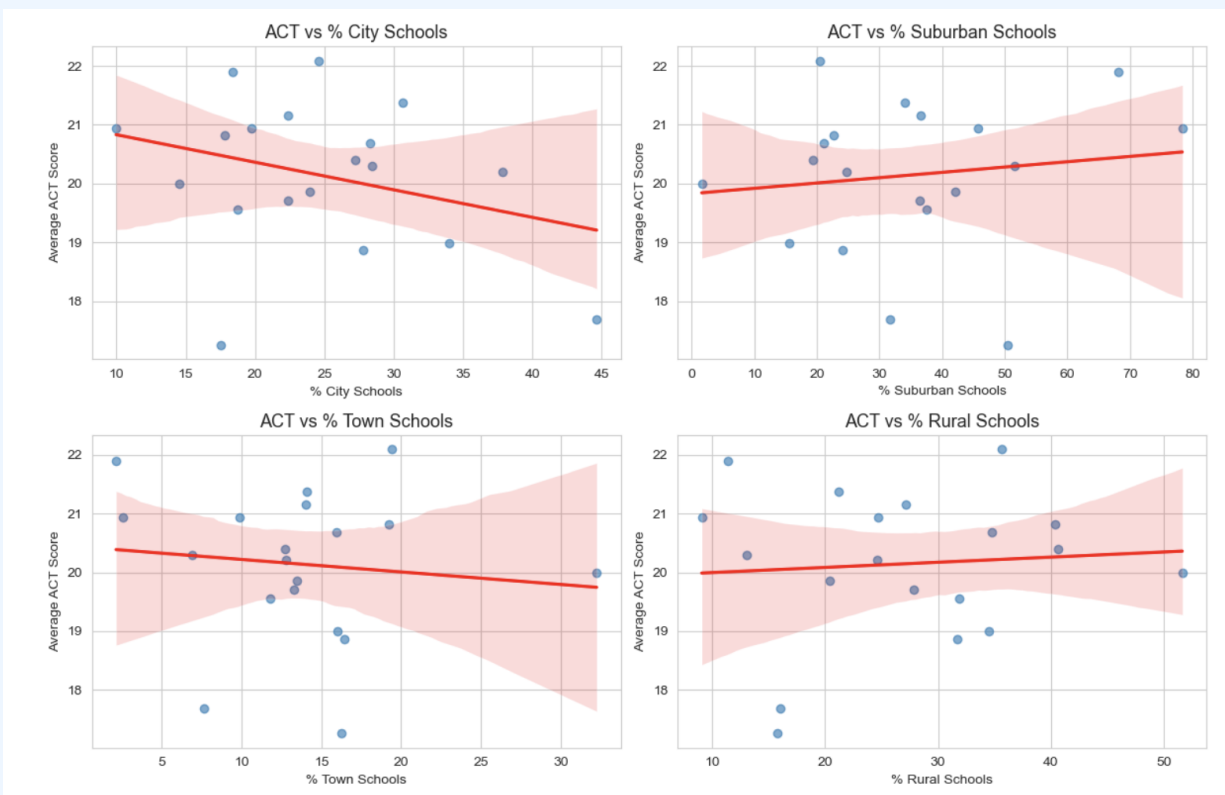


The above graphs focus on each **states average ACT/SAT exam score** while using the size of the plotted dot to represent the level of each geoassignment representation. The average ACT/SAT values remain the same across all the 4 graphs, while only the size of the dot changes across the graphs.

If we were to look at all the states that had an average ACT score below 18, we would want to focus on DE and NY. We can observe that the largest dot for Delaware (DE) is in the Suburban graph, which is telling us majority of public schools in Delaware are in a Suburban geoassignment. For the state of New York (NY), we can observe that the largest dot is in the City graph, telling us that majority of public schools in New York are in a City geoassignment.

Based off this small example, we can see how this visual makes it hard for us to determine if school geographical assignment is responsible for the average ACT/SAT exam score being that low for those two states.

Another, and more effective way to analyze this relationship is to focus on the **trend of exam scores per category**, instead of the state as the main point of interest:



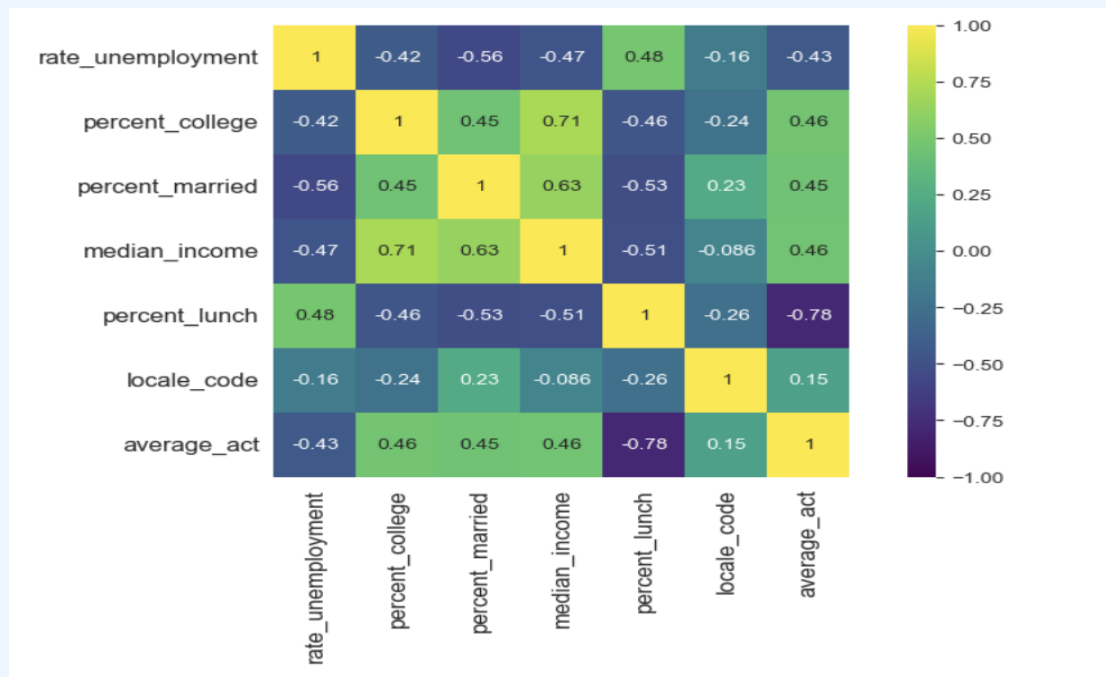
These observations suggest that Average ACT/SAT exam scores have a **stronger correlation** with states that have higher levels of Suburban schools, and a **lower correlation** with states that have higher percentages on City schools.

Exploratory Analysis, by other socioeconomic variables

Now, we want to when analyze the relationships between average **ACT/SAT exam scores for all other variables** - unemployment rate, percentage of adults with a college degree (higher-educational attainment), percentage of children in a married couple family, median household income, and percentage of students at the school that are eligible for free or reduced price lunch (poverty rate).

We started off by first plotting the correlation of numerical variables in the data to explore relationships between them. By choosing to display the relationships between the variables in a **pair plot**, we can see the direction these relationships trend, while the additional regression line helps put an emphasis on that direction.

We better analyzed these correlations by creating a **color grid** (heatmap) where each square represents one pair of variables:



The colors show the strength and direction of each correlation between the variables. yellow/green, represents a strong positive relationship, dark purple represents a strong negative relationship, and blue/green represents little to no relationship.

Based on these visuals we can determine that there is a **positive correlation** with higher-educational attainment (percent_college), married-couple households (percent_married), and median income (median_income). While there also appears to be a **negative correlation** with unemployment rate (rate_unemployment) and poverty levels (percent_lunch).

Regression Analysis

The exploratory data analysis done before now is enough to suggest patterns in our data, but regression analysis will help us quantify these patterns statistically.

We did this by printing out **two Model Fit Summaries** - one that ran a multiple regression model for the original socioeconomic variables, and another that ran the model for all variables (original 5 socioeconomic variables and geographical assignment).

When we ran the Multiple Regression Model to examine whether the socioeconomic characteristics, **outside of geoassignment**, predict average ACT/SAT exam performance, the model explained the following:

- A 62.5% of the variance in ACT scores ($R^2 = 0.625$, $p < 0.001$).
- Percent of students eligible for free/reduced lunch (poverty levels) was a **strong negative predictor** ($\beta = -7.67$, $p < 0.001$), while percent of adults with college degrees (higher-educational attainment) was a **strong positive predictor** ($\beta = 1.87$, $p < 0.001$).
- Median income and percent of children in married-couple families were **not statistically significant**.

These results suggest that higher-educational attainment and poverty levels are the strongest socioeconomic correlates of school performance.

After running the regression model for the original socioeconomic factors **with the addition of school geography assignment** (city, suburban, town, rural), the regression model output shows **little additional variation in ACT performance** ($R^2 = 0.625$). Poverty rates (percent_lunch, $\beta = -7.74$, $p < 0.001$) and higher-educational attainment (percent_college, $\beta = 1.77$, $p < 0.001$) still represent the strongest predictors of school performance.

CONCLUSION

Based on the differing average exam scores for each geoassignment category, it is safe to assume that a school's geographical assignment **does** influence school performance. But based on how little these averages differ, we cannot confidently support the claim that geoassignment necessarily has a **strong** effect on school performance.

When analyzing the relationship between school performance and all socioeconomic variables, it appears that there are **other school variables that have stronger effects on school performance** than geographical assignment. Specifically - higher-educational attainment (percent_college) and poverty levels (percent_lunch).

This was observed in our descriptive analysis but also proven statistically when we ran a Regression Model to test whether these socioeconomic variables, predicted together with geographical assignment, could significantly explain the average ACT/SAT exam scores. The results concluded that the model is **highly significant** when testing **without geographical assignment**, meaning that at least one of the original predictor variables strongly influence ACT/SAT exam scores. Specifically, higher-educational attainment (percent_college) and poverty levels (percent_lunch).

When running the regression model for the original original socioeconomic factors **with the addition of school geography assignment** (city, suburban, town, rural), the regression model output shows little additional variation in ACT performance ($R^2 = 0.625$).

More specifically, rural schools show a small but statistically significant disadvantage ($\beta = -0.13$, $p = 0.036$), while suburban and town schools are not significantly different from city schools. On the other hand, poverty rates (percent_lunch, $\beta = -7.74$, $p < 0.001$) and higher-educational attainment (percent_college, $\beta = 1.77$, $p < 0.001$) still represent the strongest predictors of school performance.

CITATIONS

- **EdGap Data:** EdGap.org. (2017). *Education gap data (EdGap_data.xlsx)* [Data set]. EdGap.
 - *Source note:* Socioeconomic data derived from the U.S. Census Bureau's American Community Survey. ACT/SAT data obtained from state departments of education and other public releases.
- **School Information Data (NCES CCD):** National Center for Education Statistics. (2017). *Common Core of Data (CCD): Public School Universe Survey, 2016–2017 (ccd_sch_029_1617_w_1a_11212017.csv)* [Data set]. U.S. Department of Education, Institute of Education Sciences. <https://nces.ed.gov/ccd/>
- **Geographical Information Data (NCES EDGE):** National Center for Education Statistics. (2017). *EDGE Geocode Public School File, 2016–2017 (EDGE_GEOCODE_PUBLICSCH_1617.xlsx)* [Data set]. U.S. Department of Education, Institute of Education Sciences. <https://nces.ed.gov/programs/edge/>
- **Geographical Assignment Reference:** National Center for Education Statistics. (2020). *EDGE Geocode Public School Technical Documentation (EDGE_GEOCODE_PUBLIC_TECHDOC.pdf)*. U.S. Department of Education. https://nces.ed.gov/programs/edge/docs/EDGE_GEOCODE_PUBLIC_TECHDOC.pdf